Firm Structure and Occupational Sorting

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[Latest Version]

The allocation of workers across occupations is a key aspect of the labor market, and it is shaped by firms' internal organization. We document a novel empirical pattern of sorting between workers and multi-worker firms using administrative data from Germany. Looking at job switchers, we find that workers that switch to higher paying firms, on average, find themselves with more subordinates on the firm wage distribution, but at a lower relative rank within the firm. These patterns imply negative assortative matching across layers but positive assortative matching across ranks. We build a tractable many-to-one assignment model with endogenous firm structure choice that allow us to tackle to question of how structure choice shapes sorting. Firms choose how many layers they want to organize their production in, so each job is indexed by a pair of firm productivity and its position in the firm hierarchy. The model rationalizes the evidence by generating negative assortative matching across layers but positive assortative matching across ranks and delivers a parsimonious way to compare jobs across firms. We provide a sufficient condition for positive sorting, namely that task importance at higher layers cannot rise too steeply relative to lower layers. The framework nests extensions with peer effects and search frictions while matching additional moments on wage dispersion.

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FIRM STRUCTURE AND OCCUPATIONAL SORTING

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MOTIVATION

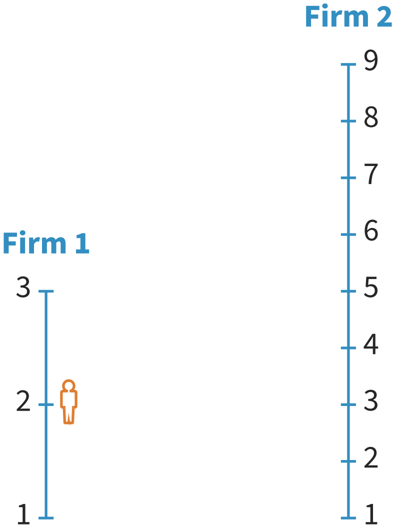
- Allocation of workers across and inside firms is key for productivity
- Workers sort into occupations/hierarchies as well as firms
- However, occupational/hierarchical structure is endogenous to labor market conditions

How does hierarchical structure affect the sorting patterns we expect to see across occupations and firms?

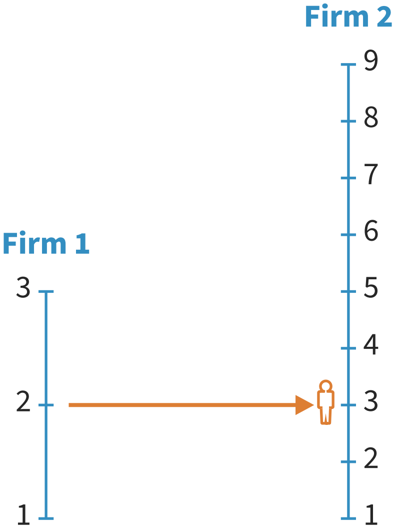
WHY STRUCTURE MATTERS

- Firm structure reflects firm productivity/quality of projects
- By selecting into firms, workers also select into occupations
- We expect to see some correlation between where a person works and their position in a hierarchy
- But how to discipline that?

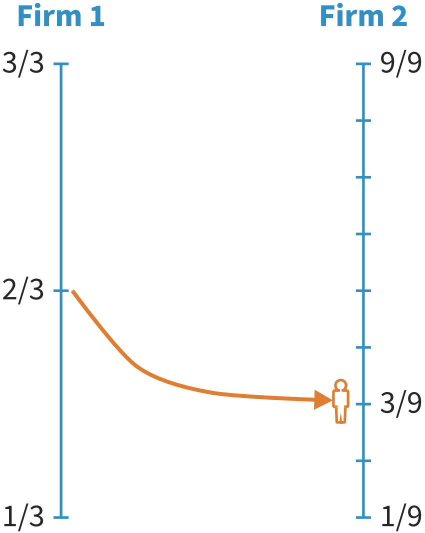
WHY STRUCTURE MATTERS



WHY STRUCTURE MATTERS



WHY STRUCTURE MATTERS



THIS PAPER

- Data - Stylized facts
  - German matched employer-employee data (LIAB); entire composition of firms
  - Job switchers when moving to better firms (higher leave-one out avg wage)
    - ★ Weakly higher number of subordinates
    - ★ Strongly *smaller* relative rank in the hierarchy
- Theory - Model of firm structure and sorting
  - Tractable setting: endogenous production function, heterogeneous workers inside the firm, and nontrivial sorting patterns
  - Comparing hierarchies: NAM across *layers* but PAM across *ranks*
  - Aggregation result that reconciles this model with canonical macro models

LITERATURE AND CONTRIBUTION

- **Sorting with multi-worker firms** (Kremer (1993), Kremer and Maskin (1996), Eeckhout and Kircher (2018), Boerma, Tsyvinski, and Zimin (2021))
  - Endogenous firm size, heterogeneous workers
- **Knowledge hierarchies** (Garicano (2000), Garicano and Rossi-Hansberg (2006))
  - Richer sorting patterns, with NAM among occupations
- **Endogenous firm structure** (Deming (2017), Adenbaum (2022), Freund (2022))

EMPIRICAL MOTIVATION

LIAB - WORKER-ESTABLISHMENT DATA

- Entire biographies of the complete workforce of a sample of establishments
- Organized in a yearly panel 2010-2017 (Dauth and Eppelsheimer (2020)) ◂ Cleaning
- Information on wages, occupation (KldB 2010), establishment identifier, industry, location, etc.
- Focus on West Germany, private sector, full-time workers aged 20-65
- Able to identify job switchers and compare its position in the firm before and after the switch

LAYERS AND RANKS WITHIN FIRMS

- Order workers within each firm-year by real wage
- Define a layer a worker is as number of workers with lower wages (can also work with binned layers)
- Measure of how many "subordinates" a worker has
- Define rank as the relative position in the hierarchy (rank = layer / firm size)
- Measure of how close to the top of the hierarchy a worker is

JOB SWITCHERS

- Focus on workers that switch firms from one year to the next (Jarosch, Oberfield, and Rossi-Hansberg (2021),Gregory (2020))
- Measure of firm quality: leave-one-out average wage
- How does a change in firm quality affect layers and ranks?
- For a worker i moving from firm j to j' at time t:

$$log(\phi_{ij't}) - log(\phi_{ijt-1}) = \beta_0 + \beta_1 \left[log(z_{j't}) - log(z_{jt-1})\right] + \text{Controls} + \varepsilon_{ijt}$$

$$\phi_{ijt} \in \{\text{Layer}_{ijt}, \text{Rank}_{ijt}\} \text{ and } z_{jt} = \text{quality of firm } j \text{ at} t$$

- Controls: Firm size, age, tenure, occupation, industry, location, year fixed effects

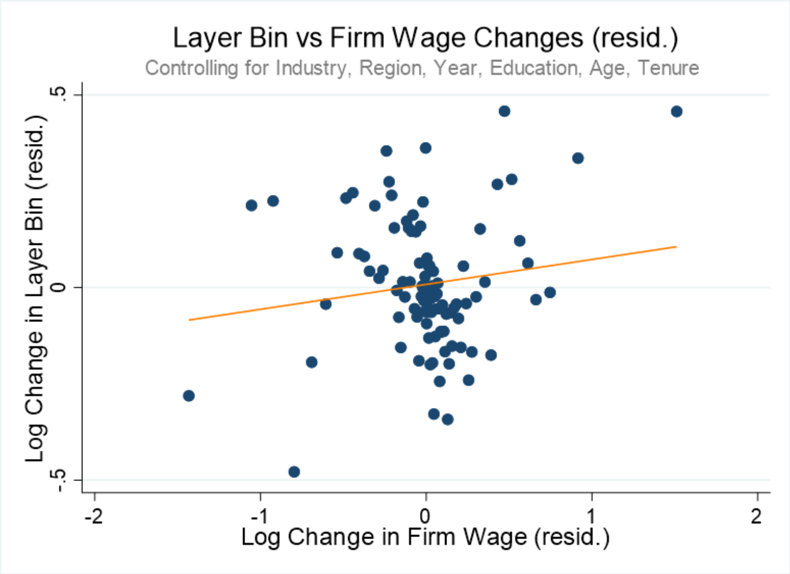
JOB SWITCHERS

Job Switch: Effect of Firm Quality Changes on Rank, Layer, and Layer Bin

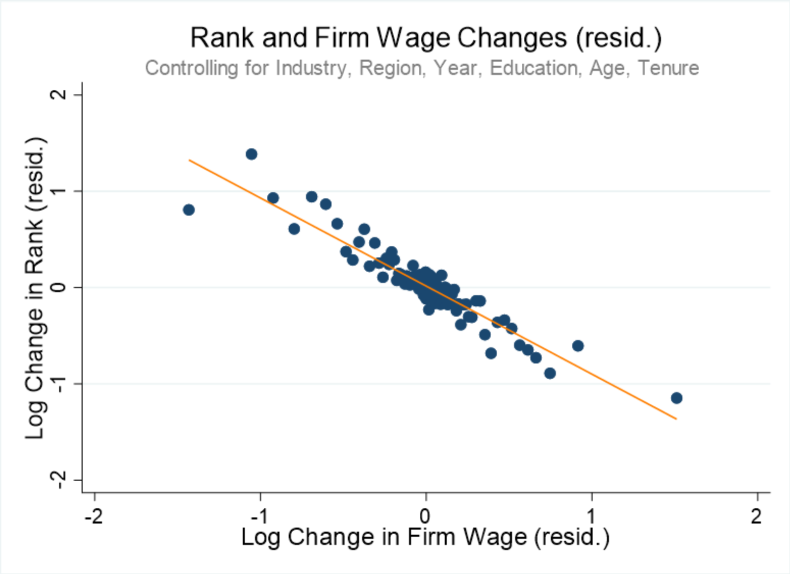
	Layer	Layer Bin	Rank
Log Diff z	0.143	0.065	-0.915
(p-value)	(0.471)	(0.473)	(0.000)
N	7,177	7,177	7,177

- Layers: we are rejecting PAM; if anything increase in firm quality leads to weakly a higher layer (NAM)
- Ranks: evidence of PAM: Increase in 10% in firm quality leads to decrease in rank of 9.15%

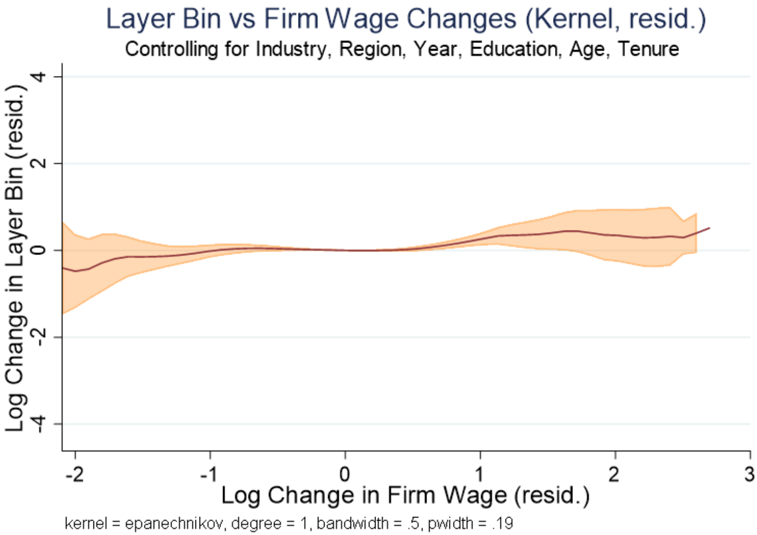
RESIDUALIZED BINSCATTER PLOTS



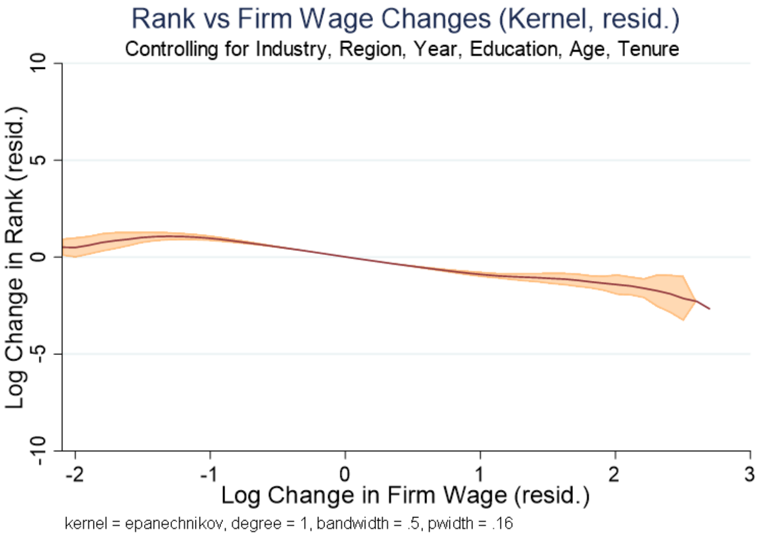
RESIDUALIZED BINSCATTER PLOTS



KERNEL REGRESSIONS - REJECT PAM ON LAYERS



KERNEL REGRESSIONS - EVIDENCE OF PAM ON RANKS



SUMMARY OF RESULTS

- Interesting sorting patters of workers and multi-worker firms
- Layers: If anything, better firms allocate workers in higher layers than worse firms
  - Better firm, the marginal contribution of each layer is smaller
  - NAM across layers
- Ranks: Better firms tend to allocate workers in lower ranks than worse firms
  - PAM across ranks
- Need of a framework that help us disentagle notions of layers and ranks within firms and rationalize these patterns

MODEL

MODEL SETUP

- Firms of type $z \sim G(z)$, workers of type $q \sim H(q)$
- Production organized in different layers/occupations indexed by n
- Firms choose number of layers (m) quality of worker in each layer (q_{nm})
- Workers choose (z, n) to maximize wages
- Number of layers determines the relative importance of each task for production

BASELINE MODEL

- Production function:

$$F\left(z, \{q_{nm}\}_{n=0}^{m}\right) = z \int_0^m a(n,m) q_{nm}^{\sigma} dn, \quad \sigma < 1 \tag{1}$$

- $a(n,m)$ is the relative importance of task n, $\frac{\partial a}{\partial n} > 0$, $\frac{\partial a}{\partial m} < 0$ ▸ Heuristic

FIRM'S PROBLEM

- The firm's problem consists of two stages
- Given structure choice, choose $\{q_{nm}\}_{n=0}^{m}$ to maximize operation profit:

$$\pi(z, m) = \max_{\{q_{nm}\}_{n=0}^{m}} z \int_0^m a(n, m) q_{nm}^{\sigma} dn - \int_0^m w(q_{nm}) dn \tag{2}$$

- When entering, choose m to maximize lifetime profit:

$$\max_m \pi(z, m) - c(m) \tag{3}$$

LABOR MARKET EQUILIBRIUM

- Main equilibrium objects: $w(q_{nm})$ and $\mu(z, n, m)$ ▸ Def
- FOC:

$$w'(q_{nm}) = \sigma z a(n, m) q_{nm}^{\sigma - 1}$$

- Guess (and later verify) $q_{nm} = \mu(n, m) z$

$$w(q_{nm}) = \frac{\sigma}{1 + \sigma} \frac{a(n, m) q_{nm}^{1+\sigma}}{\mu(n, m)}$$

LABOR MARKET EQUILIBRIUM

- Profit: ‣ Solution Algorithm

$$\pi(z,m) = \frac{z^{1+\sigma}}{1+\sigma}\int_0^m a(n,m)\mu(n,m)^\sigma dn$$

- To illustrate: assume $a(n,m) = \frac{a_0 n}{m}$, conjecture $\mu(n,m) = \frac{\mu_0 n}{m}$

STRUCTURE CHOICE

- Firm chooses m to maximize

$$\max_{m}\ \pi(z, m) - c(m)$$

- Assume $c(m) = \frac{\kappa m^2}{2}$, then FOC implies

$$m(z) = \beta z^{1+\sigma}$$

ALLOCATION

- Feasibility $\Rightarrow$ find $\mu_0 > 0$ that clears the market
- Allocation:

$$\mu(z,n,m) = \frac{\mu_0 n}{m(z)} z \Rightarrow \mu(z,n) = \frac{\mu_0 n}{\beta z^{\sigma}}$$

- NAM across firms (given n) comes from convexity in firm size
- Average worker inside firm $\bar{q}(z) \propto z^{2+\sigma}$ is increasing (PAM on average)

WAGES

- Plugging into $w(q_{nm})$:

$$w(z,n,m) = \frac{\sigma}{1+\sigma}\frac{a_0\mu_0^\sigma n^{1+\sigma}}{m^{1+\sigma}}z^{1+\sigma}$$

- CEO pay grows exponentially:

$$w(z,m,m) = \frac{\sigma}{1+\sigma}a_0\mu_0^\sigma z^{1+\sigma}$$

SUPERMODULARITY

- Reminder: a function $f(x, y)$ is supermodular if

$$f_{xy} > 0$$

- Sufficient condition for PAM in classical Becker (1973) model
- A function is log-supermodular if

$$\frac{f_{xy} f}{f_x f_y} > 1$$

- Strong condition, usually sufficient for PAM in search models (Shimer and Smith 2000, Eeckhout and Kircher 2011)

SUPERMODULARITY

- But production function is supermodular, shouldn't we expect PAM?
- We can write the production function as

$$F\left(z, \{q_{nm}\}_{n=0}^{m}\right) = \int_0^m a(n,m)\, f(z, q_{nm})dn$$

- Supermodularity is not enough anymore. Now we need

$$\frac{f_{zq}(z,q)\, f(z,q)}{f_z(z,q)\, f_q(z,q)} > \frac{|\varepsilon_{a,m}||\varepsilon_{m,z}|}{\varepsilon_{f,z}}$$

- Potentially stronger than log-supermodularity!

SUPERMODULARITY

- Convexity in structure choice makes it so importance dominates gains in productivity
- Can show that under certain conditions this boils down to

$$\left|\varepsilon_{a,m}\right| < \frac{1}{1+\sigma}$$

▸ Proposition

- Idea: $a(n, m)$ is "concave enough"

COMPARING HIERARCHIES

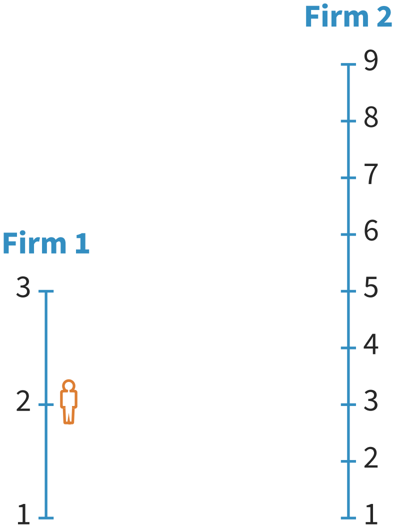
- When would two firms hire the same worker?

$$\mu(z, n) = \mu(z', n') \Leftrightarrow \frac{n'}{n} = \left(\frac{z'}{z}\right)^{\sigma}$$

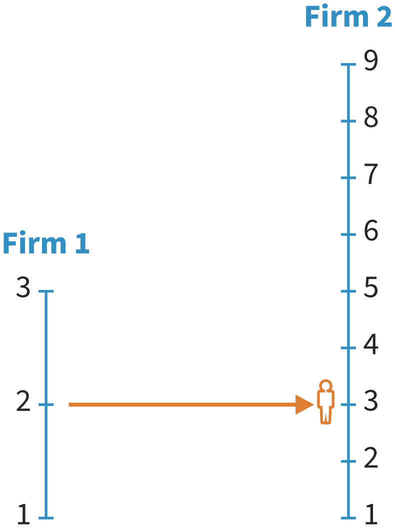
- Define $\varphi = \frac{n}{m}$ (worker rank in the firm):

$$\mu(z, n, m) = \mu(z', n', m') \Leftrightarrow \frac{\varphi}{\varphi'} = \frac{z'}{z}$$

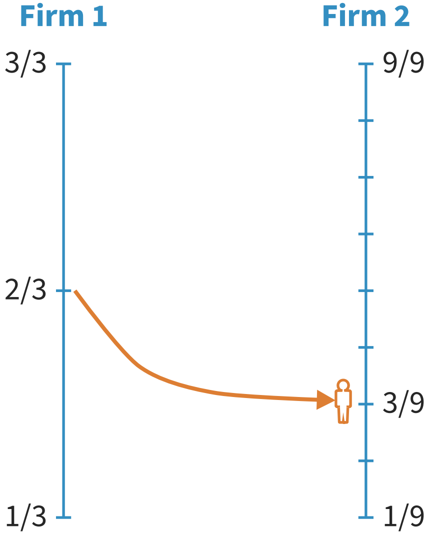
COMPARING HIERARCHIES



COMPARING HIERARCHIES



COMPARING HIERARCHIES



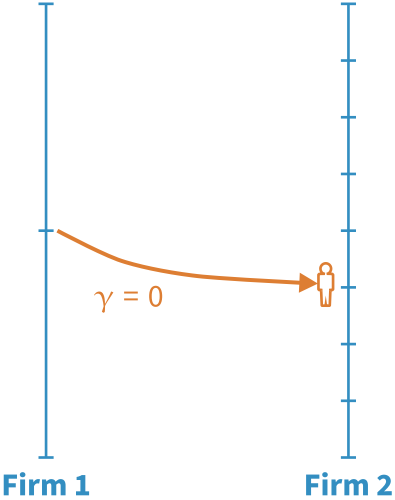
COMPARING HIERARCHIES: DISCUSSION

- Recontextualizes job transitions
- What is the correct notion of PAM in this case?
- Suppose we have peer effects (similar to Manski's reflection problem in logs):

$$f(z, q_{nm}) = z \left(\int_0^m q_{nm} dn \right)^{\gamma} q_{nm}^{\sigma}$$

  - Stronger peer effects ($\uparrow \gamma$) exacerbate "PAM-ness" in ranks

COMPARING HIERARCHIES (PEER EFFECTS)



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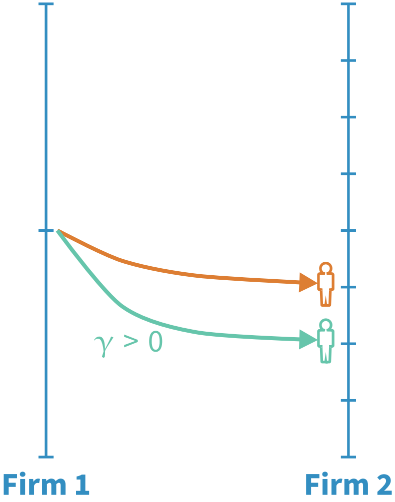
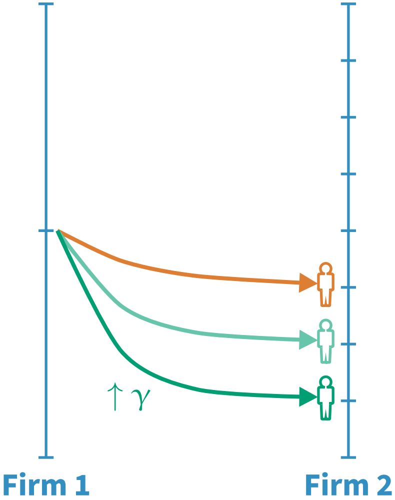


COMPARING HIERARCHIES (PEER EFFECTS)



POWER LAWS

- Suppose z follows a power law

$$Prob(z > x) \sim S(x)x^{1-\eta}, \quad \eta > 1$$

Proposition

If z follows a power law, then $m(z)$ also follows a power law. In particular, if z is Pareto with scale parameter z_{min} and shape parameter $\eta > 1 + \sigma$, then $m(z)$ is Pareto with scale $\beta z_{min}^{1+\sigma}$ and shape $\frac{\eta}{1+\sigma}$.

- Under Pareto assumption, can recover distribution of z from hierarchies.

POWER LAWS

- Important fact from wage dispersion literature: right tail of distribution of log wages is convex

Proposition

If the right tail of q follows a power law with $S(x) = s$, then the distribution of log wages is convex

- Disciplines the types of distributions that are reasonable for q

WAGE DISPERSION

- Maintaining Pareto assumption, we can compute within-firm log wage dispersion:

$$V^{within} = \mathbb{E}\left[Var\left(\log w|z\right)\right]$$

- Between-firm log wage dispersion:

$$V^{between} = Var\left(\mathbb{E}\left[\log w|z\right]\right)$$

- Use the model to target relative sizes of within vs. between firm log wage dispersion
- Illustration: Song et al. (2019) estimate a ratio of $\frac{V^{within}}{V^{between}} \approx 1.5$

WAGE DISPERSION

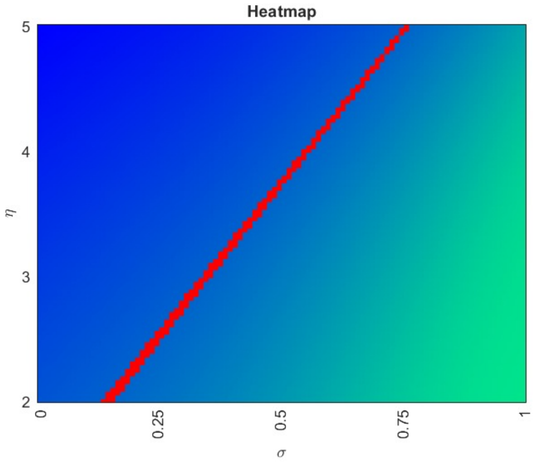
$\kappa = 0.1$

$z_{min} = 1$

Green is high

Blue is low

Red means $\frac{V^{within}}{V^{between}} \approx 1.5$



$a_0 = 1$

AGGREGATION

- Aggregate output

$$Y = \int F\left(z, \{\mu(z,n,m)\}_{n=0}^{m}\right) dG(z)$$

Proposition

Suppose z follows a Pareto distribution with minimum value z_{min} and the production function follows (1). Then $Y = AL$, where L is the total effective units of labor in the economy and A is a constant.

EXTENSION: DIRECTED SEARCH

- Suppose firms post wages and workers directedly search for jobs
- Define $\lambda(z, n, m)$ as the tightness ratio and $p(\lambda)$ is the prob of finding a job
- Firms solve

$$\max \int_0^m p(\lambda(z, n, m)) \left(a(n, m) f(z, q_{nm}) - w(q_{nm})\right)$$

- Workers solve

$$\max_{(z,n,m)} \frac{p(\lambda(z, n, m))}{\lambda(z, n, m)} w(q_{nm})$$

EXTENSION: DIRECTED SEARCH

- Similar environment to Eeckhout and Kircher (2011)
- However, supermodularity requirement is potentially even stronger now

$$\frac{f_{zq}(z,q)\,f(z,q)}{f_z(z,q)\,f_q(z,q)} > \frac{|\varepsilon_{a,m}||\varepsilon_{m,z}| + |\varepsilon_{p,\lambda}||\varepsilon_{\lambda,z}|}{\varepsilon_{f,z}}$$

- Can't find conditions on primitives here yet, but may be promising

FINAL REMARKS

- Empirical evidence of rich patterns of sorting between workers and multi-worker firms
- Constructed a sorting model with multi-worker firms and structure choice has empirically consistentconsequences for sorting
- The importance function + assignment function give us a way of comparing workers/occupations across firms of different productivities

NEXT STEPS

- Characterize the directed search extension and quantitative analysis
- Provide an estimate for the importance and assignment function using model and data
- Explore implications for firm dynamics and growth
- Counterfactuals: …

Thank You!

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Appendix

MICROFOUNDATION

- Law firm with two workers, lawyer and paralegal, each with 1/3 units of time
- Cases have a random "profile" $x \sim \mathcal{U}\,[0, 1]$
- The firm chooses whether to accept the cases and to whom to assign the case
- Solution: accept cases with profile $x \geq 1/3$
- Assign cases $x \in [1/3, 2/3]$ to paralegal, $x \geq 2/3$ to lawyer
- Avg productivity of paralegal = 1/2, avg productivity of lawyer = 5/6

Back

MICROFOUNDATION

- Now firm has hired a senior lawyer
- What happens to the others' participation in total revenue?
- New solution: accept all cases, $x \leq 1/3$ to paralegal, $x \in [1/3, 2/3]$ to lawyer, $x \geq 2/3$ to senior lawyer
- Paralegal participation: $1/2 \rightarrow 1/6$, lawyer participation: $5/6 \rightarrow 1/2$
- Firm hires senior lawyer if additional caseload is worth it

▸ Back

DEFINITION OF EQUILIBRIUM

Definition

A competitive equilibrium is a set of functions $\mu : \mathcal{Z} \times \mathbb{R} \times \mathbb{R} \to \mathcal{Q}$, $w : \mathbb{Q} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$, and $m : \mathcal{Z} \to \mathbb{R}$ such that

i. μ solves (2)

ii. m satisfies (3)

iii. w satisfies worker optimality condition:

$$w(q, n, m) \geq w(q, n', m'), \;\; \forall (n', m') \;\text{ s.t. }\; n' \leq m' = m(z), \;\text{ for some }\; z \in \mathcal{Z}$$

iv. the allocation is feasible:

$$\int_{z}^{\bar{z}} \int_{0}^{m(z)} dndG(z') = \int_{0}^{m(z)} \int_{\mu(z,n,m)}^{\bar{q}} dH(q')dn, \;\; \forall z \in \mathcal{Z}$$

Back

SOLUTION ALGORITHM

- Define $\varphi \equiv \frac{n}{m}$
- Rewrite maximization problem independent from m

$$\max_{\{q_\varphi\}_{\varphi=0}^{1}} \int_0^1 a(\varphi)\, f(z, q_\varphi) d\varphi - \int_0^1 w(q_\varphi)$$

- FOC implies

$$w'(\mu(z, \varphi)) = a(\varphi)\, f_q(z, \mu(z, \varphi))$$

SOLUTION ALGORITHM

- Feasibility:

$$\int_{z}^{\bar{z}} \int_{0}^{\varphi} g(z')d\varphi' dz' = \int_{\mu(z,\varphi)}^{\bar{q}} h(q')dq', \ \forall z$$

- This implies the following differential equation

$$\frac{\partial \mu(z,\varphi)}{\partial z} = \varphi \frac{g(z)}{h(\mu(z,\varphi))}$$

SOLUTION ALGORITHM

- Then, labor market equilibrium is given by a family of functions $\{w_\varphi, \mu_\varphi\}_{\varphi=0}^{1}$ that satisfy the following system of differential equations:

$$\frac{dw_\varphi(z)}{dz} = a(\varphi)\, f_q(z, \mu_\varphi(z)) \frac{d\mu_\varphi}{dz}(z)$$

$$\frac{d\mu_\varphi(z)}{dz} = \varphi \frac{g(z)}{h(\mu_\varphi(z))}$$

Back

SUPERMODULARITY PROPOSITION

Proposition

Suppose $a(n, m) = \tilde{a}(\frac{n}{m})$, where $\tilde{a}(\cdot)$ is a polynomial that takes $\frac{n}{m}$ as its argument. Then, the allocation exhibits PAM if

$$\left|\varepsilon_{a,m}\right| < \frac{1}{1+\sigma}$$

▸ Back